Criterion 3: Outcome Based Assessment (120)

3.1 Evaluation of Continuous Assessment (10)

(Describe the process of evaluation followed during continuous assessment to maintain quality of assessment; constructive alignment of questions with COs and for evaluation.)

3.1.1 Continuous Assessment Process

The Department of Computer Science and Engineering at Dr. B. C. Roy Engineering College follows a structured Continuous Internal Evaluation (CIE) process in a examination system prescribed by Maulana Abul Kalam Azad University of Technology (MAKAUT), West Bengal. Since the institute is affiliated to MAKAUT and t the curriculum structure, assessment scheme, and semester-end examination process are governed by the affiliating university.

The Continuous Assessment process is designed to:

- Evaluate student learning continuously throughout the semester,
- Ensure attainment of Course Outcomes (COs),
- Establish constructive alignment among COs, POs and PSOs,
- Monitor academic progress of students, and
- Support continuous quality improvement in teaching-learning practices.

The department follows both direct and indirect assessment mechanisms. However, Course Outcome attainment is primarily measured using direct assessment too

The assessment process is monitored by:

- Course Coordinator,
- Department Academic Committee (DAC),

The direct assessment process includes:

- Continuous Assessments (CA1, CA2, CA3 and CA4),
- Practical Continuous Assessments (PCA1 and PCA2),
- Assignments,
- Quizzes,
- Tutorials,
- Viva-Voce,
- Coding exercises,
- Laboratory experiments,
- Seminar presentations,
- Mini-projects,
- Technical report writing, and
- Semester End Examinations (SEE).

3.1.1(a) Theory Course Continuous Assessment Process

For theory courses, the assessment process consists of:

- Four Continuous Assessments (CA1, CA2, CA3 and CA4), and
- Semester End Examination (SEE).

The assessment structure prescribed by MAKAUT is shown below.

Table 3.1.1(a): Theory Course Assessment Structure

Assessment Component	Conducting Authority	Marks **
CA1	Department	25
CA2	Department	25
CA3	Department	25
CA4 (Online MCQ Based)	MAKAUT University	25
Semester End Examination (SEE)	MAKAUT University	70

**5 marks awarded based on the attendance

Note: Additional attendance marks are awarded as per university regulations.

The internal continuous assessments (CA1, CA2 and CA3) are conducted by the department during the semester through suited for individual courses:

- class tests,
- assignments,
- tutorials,
- quizzes,
- coding exercises,
- viva-voce,
- presentations, and
- report writing.

The CA4 examination is conducted centrally by MAKAUT in online MCQ mode.

The Semester End Examination (SEE) question papers are prepared, moderated, and evaluated by MAKAUT University.

3.1.1(b) Practical Course Assessment Process

For laboratory/practical courses, all assessments are conducted internally by the department following university guidelines.

Practical assessment process consists of:

- Practical Continuous Assessment-1 (PCA1),
- Practical Continuous Assessment-2 (PCA2), and
- End Semester Practical Examination.

Table 3.1.1(b): Practical Course Assessment Structure

Assessment Component	Conducting Authority	Marks
PCA1	Department	40
PCA2	Department	40
End Semester Practical Examination	Department/Internal-External Examiners	60

The practical evaluation process includes:

- Experiment execution,
- Coding/implementation,
- Observation and analysis,
- Laboratory notebook evaluation,
- Viva-voce, and
- Mini-project/demo wherever applicable.

Laboratory manuals are prepared in alignment with the syllabus prescribed by MAKAUT and are made available to students before commencement of the laboratory sessions.

3.1.2 Continuous Assessment Process Flow

Course Allocation
↓
Preparation of Course File
↓
Definition/correction (if required) of COs and CO-PO/PSO Mapping
↓
Preparation of Lesson Plan and Assessment Plan
↓
Preparation of Question Papers and Assignments
↓
Question Paper Moderation
↓
Conduct of Continuous Assessments
↓
Evaluation using Rubrics and Scheme
↓
CO-wise Attainment Calculation
↓
Corrective Actions and Remedial Measures

3.1.3 Quality Assurance Practices in Continuous Assessment

Table 3.1.3(a): Quality Assurance Practices

Sl. No	Quality Practice	Description
1	CO-based Assessment	Questions mapped with Course Outcomes
2	Bloom's Taxonomy Mapping	Questions framed at different cognitive levels
3	Internal Moderation	Question papers reviewed by senior faculty
4	Transparent Evaluation	Evaluated scripts discussed with students
5	Rubric-Based Assessment	Standard rubrics used for labs, projects and sessional courses
6	Continuous Monitoring	Performance monitored by DAC and HoD
7	Corrective Measures	Remedial classes for weak students

3.2 Evaluation of the Semester End Exam (SEE) Question Paper (10)

3.2 Evaluation of Semester End Examination (SEE) Question Paper (10)

The Department of Computer Science and Engineering follows the Semester End Examination (SEE) system prescribed by MAKAUT. Since the institute operates under the T examination pattern, question paper setting, moderation, and evaluation processes for Semester End Examinations are governed centrally by the affiliating university.

For theory courses, MAKAUT conducts:

- Semester End Examination (SEE) for 70 marks, and
- CA4 examination in online MCQ mode for 25 marks.

The department conducts CA1, CA2 and CA3 internally as part of the continuous evaluation process.

Faculty members of the department actively contribute to the university examination system as:

- question paper setters,
- moderators,
- examiners,
- subject experts, and
- members of academic committees whenever nominated by the university.

Faculty members prepare and submit question papers to the university following the syllabus structure and guidelines prescribed by MAKAUT. The university subsequently underta

- syllabus coverage,
- appropriate difficulty level,
- uniformity in standards,
- elimination of ambiguity,
- fairness and transparency.

3.3 Evaluation of Laboratory Work and Workshop (Continuous and SEE) (10)

3.3 Evaluation of Laboratory Work and Workshop (Continuous and SEE) (10)

Laboratory courses play a significant role in providing practical exposure, analytical ability, programming skill, problem-solving capability, and hands-on learning experience to stude

The department follows continuous and semester-end laboratory evaluation processes aligned with Course Outcomes, Program Outcomes, and Program Specific Outcomes.

3.3.1 Laboratory Evaluation Process

Preparation of Laboratory Manual
↓
Conduct of Laboratory Experiments
↓
Continuous Observation and Assessment
↓
PCA1 and PCA2 Evaluation
↓
Laboratory Record Verification
↓
Viva-Voce Examination
↓
End Semester Practical Examination
↓
CO-wise Attainment Analysis

3.3.2 Laboratory Assessment Structure

Table 3.3.2(a): Laboratory Assessment Structure

Assessment Component	Conducting Authority	Marks
PCA1	Department	40
PCA2	Department	40
End Semester Practical Examination	Department/Internal-External Examiners	60

3.3.3 Rubrics Used for Laboratory Assessment

Rubric-based assessment is used for maintaining transparency, consistency, and uniformity in laboratory evaluation.

Table 3.3.3(a): Sample Laboratory Rubrics

Evaluation Parameter	Excellent (4)	Good (3)	Average (2)	Poor (1)
Understanding of Experiment	Complete understanding	Minor conceptual gaps	Partial understanding	Poor understanding
Coding/Implementation	Error free execution	Minor errors	Moderate errors	Unable to execute
Result Analysis	Excellent analysis	Moderate analysis	Limited analysis	No analysis
Viva Performance	Excellent response	Good response	Average response	Poor response
Documentation	Complete and neat	Minor omissions	Incomplete	Poor documentation

3.3.4 Laboratory Evaluation Weightage

Table 3.3.4(a): Laboratory Evaluation Components

Component	Weightage
Experiment Execution	30%
Viva-Voce	20%
Observation and Analysis	20%
Laboratory Record	20%
Attendance and Discipline	10%

3.4 Evaluation of Industrial Training/ Internship (Continuous and SEE) (10)